

Design and Implementation of Steady State Visual Evoked Potential Based Brain Computer Interface Video Game

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Abstract—Brain Computer Interface (BCI) is a system that enables device control by processing EEG signals acquired from scalp. This system can be used as a communication channel for severely disabled patients as well as other applications for healthy people. In this study, a BCI game was implemented based on classifying steady state visual evoked potentials (SSVEP) that occur in occipital lobe when a subject is exposed to a visual stimulus flickering at a certain frequency using canonical correlation analysis (CCA). In the experiment, the subjects were expected to move a square to specific locations using four visual stimuli with different frequencies. This task could be completed at 1.4 seconds in the best case scenario. The subjects completed this task at 1.4 seconds in the best run, 36.22 seconds in the worst run, and 8.57 seconds in average. The ratio of moves towards target to all moves were calculated as 66.72%.

Index Terms—brain computer interface, steady state visual evoked potential, EEG, canonical correlation analysis.

I. INTRODUCTION

Brain Computer Interface (BCI) is a system that enables humans to control a device using their brain activity. BCI generally consists of a hardware for collecting electroencephalogram (EEG) signals and a software for interpreting the acquired signals.

BCIs were initially designed as a communication channel for severely disabled patients. Such patients can use a computer or a speller, or control a robotic arm via BCIs. BCIs are now used in other fields as well. For example, in medicine BCIs can be used in detection, diagnosis, treatment and prevention of some illnesses; in education, BCIs can be used as a neurofeedback mechanism to determine the degree of clearness of studied information; BCIs and internet of things are integrated to enable control of a smart environment [1]. For entertainment, BCIs enable interacting with virtual reality, or BCIs can be used as a control mechanism for videogames [2].

The majority of the BCI applications for healthy users consist of video games. There are many reasons why video games make up this majority. First of all, most of the gamers easily adopt new technology. Some of the gamers already wear headgear. Second, they know that in order to master something they have to train a lot; and they are very competitive, they

will work hard for the advantage provided by BCI. Third, some of the gamers do not hesitate to spend money on gaming peripherals. And finally, gaming companies can earn huge profits by introducing a new game element such as BCI, which is a powerful motivation to invest in research and development of BCIs [3] [4].

There are also many limitations for BCI video games to be commercially off-the-shelf. For instance, most of the BCI games are proof-of-concept. They are tested in laboratory conditions. EEG is very prone to noise, even eye blinking or muscle movement causes EEG data to be corrupted. In laboratory conditions, subjects' movements are often limited, but in real gaming environment players will often move or blink. Even in the laboratory conditions BCI performance (in terms of information transfer rate and latency) is pretty low, and in real gaming environment it is expected to get lower. In order to get better accuracy there are two steps taken that gamers would probably like to avoid; one is applying gel to head before EEG cap is worn, the other is calibrating the BCI system which takes about 5 to 20 minutes. But as the BCI systems are getting more advanced, even though with a little accuracy loss, these steps can be skipped. For roughly 20% of BCI users, the signals cannot be interpreted with sufficient accuracy to form some kind of control. This means 20% of gamers will not be able to use BCI, therefore adoption of BCI might be a problem in the gaming community. EEG signals have low information transfer rate and high latency. These characteristics prevent BCIs to be the main control mechanism for complex video games [3] [5].

Even though BCIs cannot be the main control mechanism of video games, they can be integrated in other ways. For instance while playing a game, the player can feel bored, relaxed, worried, anxious or aroused depending on the game's difficulty level and player's skills. BCIs can be used to monitor and detect the mental states of the player and arrange the difficulty level of the game accordingly. The player can consciously evoke particular brain signals by imagining limb movement or making mathematical calculations in order to control a game element. BCIs can be used for auxiliary control in games.

For example, BCIs can be used to give optional commands instead of critical commands. These optional commands can make the gameplay easier and more enjoyable, but they are not a necessity to complete the game. BCIs can also become handy as an extra control modality, especially in "induced" or "situational disability" moments, which means the moments when one wants have more skills and communication channels than traditional control mechanisms such as mouse or keyboard, to control the game environment. For example, instead of forcing player to hold multiple keys, BCI can be used for specific commands [2] [4].

In order to control a computer using brain signals, the user requires to be stimulated by some internal or external stimuli, or be in a specific state of mind such as concentrated or relaxed. The most popular modalities used in BCI systems are evoked potentials such as SSVEPs and P300, sensory-motor rhythms, non-motor cognitive tasks and hybrid signals.

SSVEP is elicited when the user is exposed to a regularly flickering object. When the user focuses on the flickering object, a SSVEP occurs in the visual cortex with the same frequency as the frequency of stimulus. The frequency of stimulus is generally chosen between 6-30 Hz. The disadvantages of SSVEP are, since only a limited number of frequencies are useful, number of controls are also limited, and the user's permanent attention to the stimulus is required. But SSVEPs require very short training (around 2 minutes), have very high information transfer rate (60-100 bits/min), and have high signal to noise ratio (SNR).

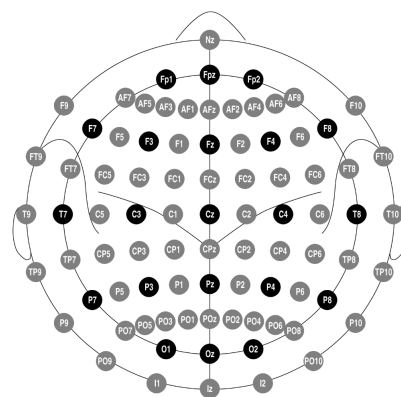
Lalor et al. [6] developed the MindBalance game, in which, the character is on a rope and the player tries to balance the character using EEG signals. There are two checkerboard images and the character is in between these images. As the character starts to lean on one side, the player can focus on the image in the opposite direction to balance the character again. Before the game was developed, some preliminary analyses were performed. 17 Hz and 20 Hz are identified as the two bilateral checkerboard frequencies that elicit the most SSVEP. For the classification, LDA was used. For the best performing feature extraction method, the average real-time control accuracy across subjects was 89%.

Van Vliet et al. [7] compared two different headsets, one is a research lab with expensive equipment, the other one is a consumer grade headset (Emotiv EPOC). The game is a tower defense game, the goal is to protect the tower from waves of enemies by building different structures and placing them on the map. When an enemy reaches the tower, the game is over. When the game starts, the user selects the places to build their defense structures. A visual stimulus is used flickering at 12.8Hz. The selections are highlighted one by one, and when a user wants to choose a selection, when it is highlighted they must look at the stimulus. Even though the accuracy obtained when using the research headset is higher compared to the accuracy obtained when EPOC headset is used, EPOC is also sufficient enough for the game, since they obtained 85% accuracy.

II. MATERIALS AND METHOD

A. Experiment Setup

EEG data were acquired using Emotiv EPOC headset. This headset has 128Hz sampling rate and 14 electrodes. The electrodes are located at AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8 and AF4 channels according to 10-10 electrode placement system as shown in Fig. 1 [8]. SSVEPs can be detected from P7, P8, O1 and O2 channels since they are located on the occipital lobe, therefore in this experiment we used the data collected from these four channels. To obtain clean data, each electrode was moisturized with a solution and we ensured that every electrode was in contact with the scalp. This process took around 10 minutes for each subject.



The data was analyzed in 3-second window four times a second using Python programming language. 3-second data is long enough for a correct classification, and short enough to enable real time gaming. The signal was classified using Canonical Correlation Analysis(CCA) and the predicted class is sent to Unity via TCP along with the correlation coefficient. If the correlation coefficient was greater than 0.3, the avatar was commanded to move according to the predicted class.

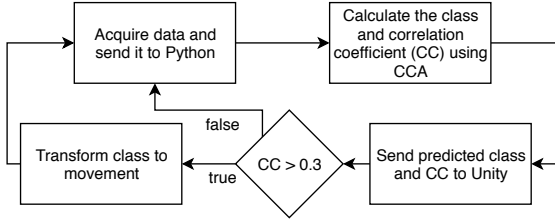


Fig. 3. Signal processing loop.

B. Visual Stimulus Color Selection

To obtain the best performance from the visual stimulus; we collected data when the subject was looking at yellow, blue, red, green, gray and white stimuli flickering at 12 Hz. To observe only the effect of the stimulus, we also collected data when the subject was not looking at any stimuli, then the background data was removed from the stimulus data. Then for all stimuli, power spectral density (PSD) was calculated using the periodogram method by using the data from O1 channel. The PSD plots are given in Figure 4.

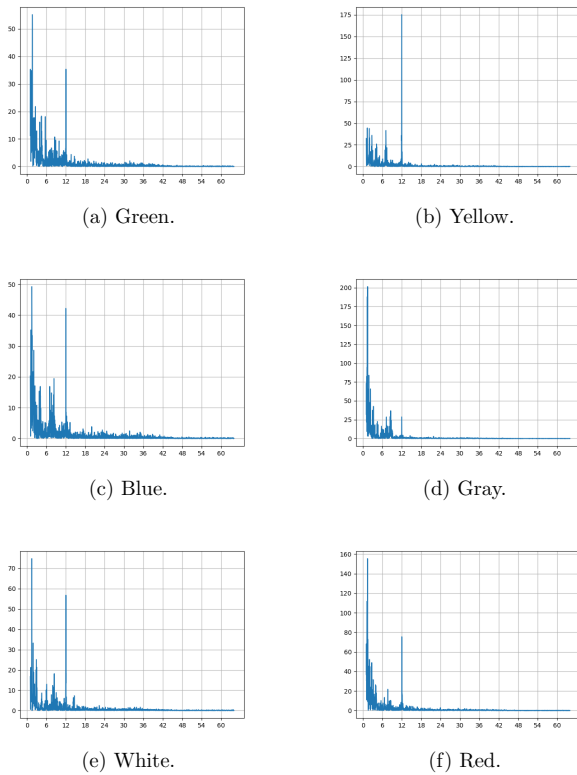


Fig. 4. PSDs emerged from different colored stimuli.

Since the stimulus frequency power is high and other frequencies' powers are low in yellow stimuli PSD, we chose the stimulus color to be yellow.

C. Canonical Correlation Analysis

CCA can be used to determine the target stimulus by using multi-channel EEG data. CCA works as follows: Let X and Y be two multidimensional variable, and $x = X^T W_x$ and $y = Y^T W_y$ be the linear combinations of these two variables. CCA finds the weight vectors W_x and W_y which maximize the correlation between x and y by solving (1).

$$\max_{W_x, W_y} \rho(x, y) = \frac{E[W_x^T X Y^T W_y]}{\sqrt{E[W_x^T X X^T W_x] E[W_y^T Y Y^T W_y]}} \quad (1)$$

In equation (1) $\max_{W_x, W_y} \rho(x, y)$ yields the maximum correlation coefficient. In SSVEP based BCIs, CCA is used to find the target stimulus in the following manner: For each possible target stimulus (in this experiment there are four possible target stimuli, flickering at 6 Hz, 6.67 Hz, 8.57 Hz and 10 Hz), a reference signal is generated using (2).

$$Y_f = \begin{pmatrix} \sin(2\pi f t) \\ \cos(2\pi f t) \\ \vdots \\ \sin(2\pi N_h f t) \\ \cos(2\pi N_h f t) \end{pmatrix} \quad (2)$$

In (2) N_h is the number of harmonics, in this experiment N_h is equal to 3. As shown in Figure 5, the correlation coefficient between each Y_f and EEG data was calculated using CCA and the stimulus that yields the maximum correlation coefficient was selected as the target stimulus [9].

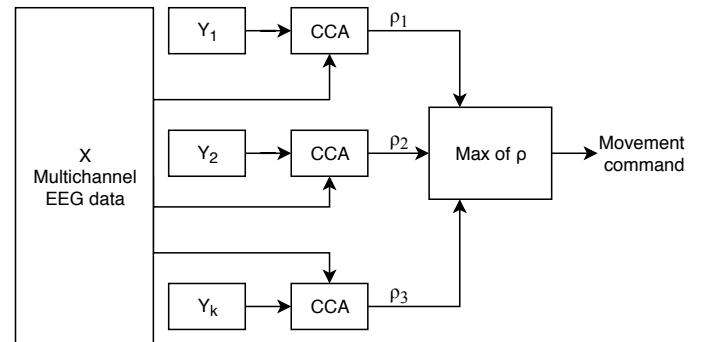


Fig. 5. CCA usage in determination of the target stimulus.

III. RESULTS

The game's performance was evaluated in two different manners. For the first evaluation, we calculated the ratio of number of correct classifications to number of all classifications. For the second evaluation, we measured the time for the avatar to eat the bait.

For the first evaluation, the distance between the avatar and the bait was calculated after every move of the avatar. If the distance was less than before, the movement was interpreted as a correct classification. If the distance was greater than before, the movement was interpreted as an incorrect classification.

The accuracies for each subject and each game play are given in Table 1. Accuracy is calculated as the ratio of number of movements towards the bait to all movements.

TABLE I
RATIO OF NUMBER OF MOVEMENTS TOWARDS THE BAIT TO ALL
MOVEMENTS (%)

	Game play 1	Game play 2	Game play 3	Average
Subject 1	66.33	66.52	66.91	66.59
Subject 2	72.66	62.23	67.90	67.60
Subject 3	68.91	68.04	72.13	69.69
Subject 4	65.98	69.45	58.99	64.81
Subject 5	60.22	60.55	68.86	63.21
Subject 6	64.67	64.92	75.64	68.41

For the second evaluation, time between the bait was generated and eaten was measured. There are two cases that extend the time for the bait to be eaten. The first one is avatar not moving due to the correlation coefficient being less than 0.3. The second one is avatar moving to the wrong direction due to incorrect classification, and then returning back to the bait.

In the best case scenario where each calculated correlation coefficient is greater than or equal to 0.3, which means each analysis becomes a command and each command is towards the correct direction, avatar takes 1.4 seconds to reach to bait. Since the game ends after avatar eats 4 baits, in the best case scenario the game takes 5.6 seconds.

The time taken for each bait to be eaten, each game play's total time and the average of time taken for each bait to be eaten are given in Table 2. The average is 8.68 seconds.

TABLE II
TIME TAKEN FOR EACH BAIT TO BE EATEN (SECONDS)

Game play	Bait	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Subject 6
1	1. bait	1.93	12.37	5.82	5.92	16.19	4.18
1	2. bait	3.95	15.12	35.16	24.23	11.55	9.10
1	3. bait	8.10	1.42	1.42	1.44	9.75	1.42
1	4. bait	1.43	13.13	11.02	6.76	1.42	8.08
1	Average	3.85	10.51	13.36	9.59	9.73	5.70
1	Total	15.41	42.04	53.42	38.36	38.90	22.78
2	1. bait	4.20	1.40	4.80	6.93	10.81	4.28
2	2. bait	10.26	36.22	15.87	5.22	13.08	6.00
2	3. bait	7.22	1.42	1.46	4.04	6.28	4.52
2	4. bait	7.90	12.98	5.64	4.75	3.22	9.30
2	Average	7.39	13	6.94	5.23	8.35	6.03
2	Total	29.58	52.02	27.77	20.94	33.40	24.10
3	1. bait	4.16	2.93	25.54	2.87	10.23	2.42
3	2. bait	3.17	7.96	15.06	12.53	9.18	9.33
3	3. bait	8.72	10.09	5.02	14.24	11.01	7.17
3	4. bait	1.44	10.85	4.57	13.78	9.38	16.58
3	Average	4.37	7.96	12.55	10.86	9.95	8.88
3	Total	17.49	31.83	50.18	43.43	39.80	35.51

IV. CONCLUSIONS

In this study, an SSVEP based BCI video game was implemented using CCA method. In the game, the player was expected to correctly choose one of the four options. The average accuracy of the subjects were found to be 66.72%. Also, the task that takes 1.4 seconds in the best case scenario took 8.57 seconds on average.

To improve the results, the threshold for the correlation coefficient can be customized for each subject according to the result of a training session. Before classification, noise removal techniques can be used such as filtering. Classification algorithms other than CCA can be employed. Other than these, the content and cosmetics of the game can be improved.

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